Full paper outline

Practical Incentive-Compatible Drafts: Carry-Over Lottery Allocation

MIT Sloan Sports Analytics Conference 2027 (SSAC27), Basketball track, Paper ID 100

Draft for the team, 3 November 2026

Deadline and format

- The paper is due at 11:59 PM EST on Tuesday, 1 December 2026, and SSAC accepts no late submissions. Prof. Highley submits it with Paper ID 100 (SSAC email, 31 October).
- SSAC requires its Word template and does not support LaTeX. The template sets no page limit, notes that papers typically run under 8 pages, allows an appendix and any mainstream citation style, and puts only the title, track and Paper ID in the title block.
- We draft in Overleaf and convert to the template with pandoc. A dry run on the current manuscript by 10 November tests the conversion.

Shape

The main text presents Carry-Over Lottery Allocation (COLA) in about 8 template pages, which is about 4,000 words with three tables and one figure. The appendix holds the proofs, the rule definitions and the full tables, and every number regenerates from the public repository.

§	Section	Words	Abstract promise it keeps
1	Introduction	500	The dilemma, the reframe, the contributions
2	Background	350	Munro and Banchio's impossibility result; 3-2-1
3	The COLA framework	600	Three dials; 3-2-1 as an edge case
4	Incentive compatibility	550	The proof and the one assumption that needs justification
5	When the assumptions fail	450	"The impact of violating model assumptions and how to mitigate"
6	Simulation design	450	Nine mechanisms, 30 leagues, 25 seasons, paired intervals
7	Results	700	Table 1; other starting positions; six more mechanisms
8	Discussion and adoption	300	"One example of many"; auction credit
9	Conclusion	100	The dilemma resolved
	Main text	4,000	

Questions to settle first

We should agree on structure and content before anything else.

1. Which parts of `main.tex` carry the argument? It runs to about 12,700 words over 26 pages, and the main text has room for about 4,000.
2. How does §5 treat a team's incentive to stay eligible for the lottery? Under Simple COLA, the team with the longest wait gains about 22 draft places by losing a first-round series. The paper can fix this with a cap on the waiting number or a reset on playoff appearance, and test the fix in the simulator, or it can disclose the incentive.
3. How do we state two abstract claims our measurements qualify? 3-2-1 still rewards losing from the play-in positions, and the incentive-compatibility proof in `main.tex` covers Classic COLA, so it needs restating for Simple COLA.

Sections

1. Introduction (500 words)

- The tradeoff between helping weak teams and the tanking it invites, and the NBA's lottery changes up to 3-2-1.
- The reframe, weakness measured by multi-season playoff results, under which losing more games buys nothing.
- The contributions: the COLA framework with 3-2-1 inside it, an incentive-compatibility proof under stated assumptions, nine mechanisms simulated on the same leagues, and Simple COLA as a one-line rule with no lottery.

Draws on `main.tex` §1 (1,929 words).

2. Background (350 words)

- Evidence that teams tank.

- Munro and Banchio's result that any mechanism based on end-of-season records can reward losing.
- 3-2-1, adopted on 28 May 2026, and earlier proposals (the Gold plan, elimination-date rules, auctions).

Draws on `main.tex` §2 (680 words).

3. The COLA framework (600 words)

- The waiting number and the three dials, which set how priority increases, how it decreases and how it maps to picks. The seven-dial parameterization moves to Appendix B.
- A one-line definition of each of the nine rules.
- 3-2-1 as an edge case, where record sets priority and the only carry-over is the bar on repeat top picks. It still pays for losing from the play-in positions, by 0.37 places from eleventh-worst and 2.02 from fourteenth-worst.

Draws on `main.tex` §3 (1,933 words).

4. Incentive compatibility (550 words)

- The theorem for Simple COLA and its assumptions, with a proof sketch in the text and the full proof in Appendix A.
- The one assumption that needs justification, that teams prefer winning games to manipulating the pool of lottery teams.
- On our runs, losing never gained a team a pick under Simple COLA across 158,325 counterfactuals.

Draws on `main.tex` §3.5, which proves the result for Classic COLA under five assumptions.

5. When the assumptions fail (450 words)

- The incentive to stay eligible, priced as the draft places the longest-waiting team gains on the lottery side of the boundary: 1.4 under the old lottery, 5.0 under 3-2-1, 9.6 under Full-lottery COLA and 22 under Simple COLA, whose boundary is a first-round loss.
- Longer rebuilds after a reset, measured as the next season's pick for a team that took a top-3 pick and missed the playoffs again: 6.6 under the old lottery, 8.6 under 3-2-1, 9.4 under Full-lottery COLA and 21.0 under Simple COLA.
- Draft-class strength, which none of the rules condition on.
- Mitigations and what each adds in complexity: a cap on the waiting number, a reset on playoff appearance, and a lottery over the top picks.

Draws on `main.tex` §4 (the pool-manipulation bound and the per-series bound for Capped COLA).

6. Simulation design (450 words)

- ZenGM Basketball-GM, 30 leagues of 25 seasons on shared seeds, and paired differences with 95% intervals.
- One measure per criterion: help for a worse record, help for a longer wait, the longest wait for a series win or top-3 pick, and the average wait for the next series win.
- Validation against the NBA over 2001-02 to 2025-26. The simulation matches the average playoff drought (2.89 seasons against 2.99), the wait for the next series win (4.65 against 4.55) and the year-to-year correlation of win percentage (0.62 against 0.60).

Draws on `main.tex` §5.1–5.2 and `testbed/harness/sim_vs_real.js`.

7. Results (700 words)

- A table of all nine rules on the four measures, with paired intervals against 3-2-1.
- A table of the reward for losing by starting position. From fifth-worst it is $+1.32$ places under the old lottery, -0.63 under 3-2-1 and -0.94 under Simple COLA; from fourteenth-worst it is $+9.96$, $+2.02$ and -3.50.
- Help, where Simple COLA cuts the longest wait for a series win or top-3 pick from 16.3 seasons under 3-2-1 to 5.5.
- Parity, where the rules are level on the wait for the next series win. Droughts end 80 to 87 times per league across the nine rules and 88 times in the NBA over the same 25 seasons, so the draft changes which teams end their droughts and barely changes how often droughts end.
- Simple COLA's one parity cost, a wait to return to the playoffs 0.18 seasons longer than 3-2-1's (95% interval 0.08 to 0.28), because 39% of its top-3 picks go to teams that lost in the first round.

Draws on `runs/frontier` (all ten arms), `abstract_table.js --table1` and `tank_counterfactual.js`. The 3-2-1 pricing and the §5 measures still need scripts in the repository.

8. Discussion and adoption (300 words)

- Choosing a rule from the family, with Simple COLA for the most help and a one-line rule, and Full-lottery COLA where parity matters more or a lottery night is wanted.
- 3-2-1 as a stopgap, with COLA as the next step.
- Auction drafts, where carry-over credits that grow with waiting and reset with success allocate the currency without rewarding a worse record.
- Limitations: one simulator, tanking measured by counterfactual, and draft-class strength.

Draws on `main.tex` *§6.*

9. Conclusion (100 words)

The abstract's Conclusion, extended to the whole family.

Appendix

A. The incentive-compatibility proof.
B. The seven dials and the definition of every rule.
C. The pool-manipulation bound.
D. Full results tables with intervals.
E. The validation table against the NBA.
F. The repository and the script behind each number.

Schedule

Date	Milestone
Friday 6 November	Structure and content agreed, leads set
Tuesday 10 November	Word conversion dry run; every number regenerates from the repository
Tuesday 17 November	Complete first draft in Overleaf
Friday 20 November	Comments back from everyone
Tuesday 24 November	Final draft in the Word template (Thanksgiving is Thursday 26 November)
Tuesday 1 December, 11:59 PM EST	Prof. Highley submits, Paper ID 100

Authors and leads

The author order is Timothy Highley, Kaushik Rajan, Tannah Duncan, Ilia Volkov and Kibur Assefa. Everyone has edit access, so change it here if you see it differently.

The proposed leads are Prof. Highley for §1, §4 and §9; Kaushik for §6, §7, the appendix tables and the Word conversion; Tannah and Ilia for §2 and the material from the original study; and Kibur for the assumptions in §4 and §5.